CS3003: IT Project Management

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Part 6: Project Controlling/Monitoring

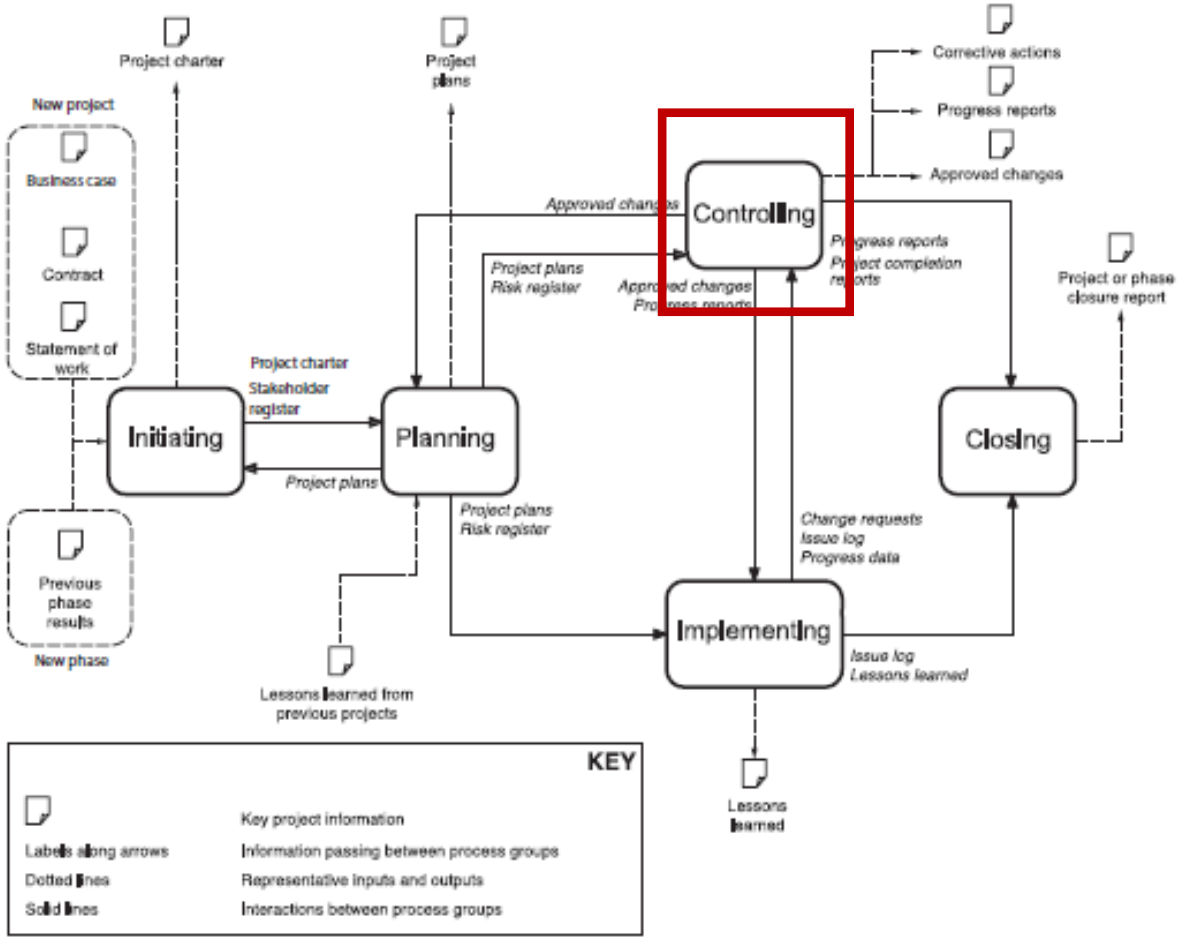
Configuration & Change Management

Control Project Progress/Performance

Project Controlling Overview and the Focus of Part 6 (see items in red)

Subject groups	Process groups: Initiating	Planning	Implementing	Controlling	Closing
Integration	4.3.2 Develop project charter	4.3.3 Develop project plans	4.3.4 Direct project work	4.3.5 Control project work 4.3.6 Control changes	4.3.7 Close project phase or project 4.3.8 Collect lessons learned
Stakeholder	4.3.9 Identify stakeholders		4.3.10 Manage stakeholders		
Scope		4.3.11 Define scope 4.3.12 Create work breakdown structure 4.3.13 Define activities		4.3.14 Control scope	
Resource	4.3.15 Establish project team	4.3.16 Estimate resources 4.3.17 Define project organization	4.3.18 Develop project team	4.3.19 Control resources 4.3.20 Manage project team	
Time		4.3.21 Sequence activities 4.3.22 Estimate activity durations 4.3.23 Develop schedule		4.3.24 Control schedule	
Cost		4.3.25 Estimate costs 4.3.26 Develop budget		4.3.27 Control costs	
Risk		4.3.28 Identify risks 4.3.29 Assess risks	4.3.30 Treat risks	4.3.31 Control risks	
Quality		4.3.32 Plan quality	4.3.33 Perform quality assurance	4.3.34 Perform quality control	
Procurement		4.3.35 Plan procurements	4.3.36 Select suppliers	4.3.37 Administer procurements	
Communication		4.3.38 Plan communications	4.3.39 Distribute information	4.3.40 Manage communications	

NOTE The purpose of this table is not to specify a chronological order for carrying out the activities. Its purpose is to map subject groups and process groups.



Configuration Management: Definition

- Configuration management (CM) is concerned with the policies, processes and tools for **controlling** and **managing assets** or **products**.
- Assets/Products include a **number of artifacts**, both **Project** and **Systems'** related, e.g. source code, executables, component libraries, files, documentation, manuals, project-related documents.
- Typical concerns: **Change Control and Versions**, **Status**, **Ownership**, **Relationships**, **Safe and Secure Storage**.
- Note that **Project-Level** CM is different to **Organization-Level** CM, however the former should be consistent with the policies and practices of the latter.

Configuration Management: Key Concepts

- **Configuration Item (CI): Any artifact** (e.g. design, code, test scenarios, document, etc.) that has been placed **under configuration control**, i.e. ensuring that versions are recorded and maintained and all versions are identified and stored for the lifetime. There are often different versions of a configuration item. Configuration items have a unique name.

  Deciding what level of configuration management will be required in a project (or in the overall organization) and planning how this level is to be achieved is an important *planning* consideration.

- **Version:** An instance of a configuration item that differs, in some way, from other instances of that item. Versions always have a unique identifier, which is often composed of the configuration item name plus a version number.

- **Version management:** Keeping track of the multiple versions of assets/products and ensuring that changes made (e.g. by different developers) do not interfere with each other.

- A **Codeline** is a set of versions of a software component and other configuration items on which that component depends.

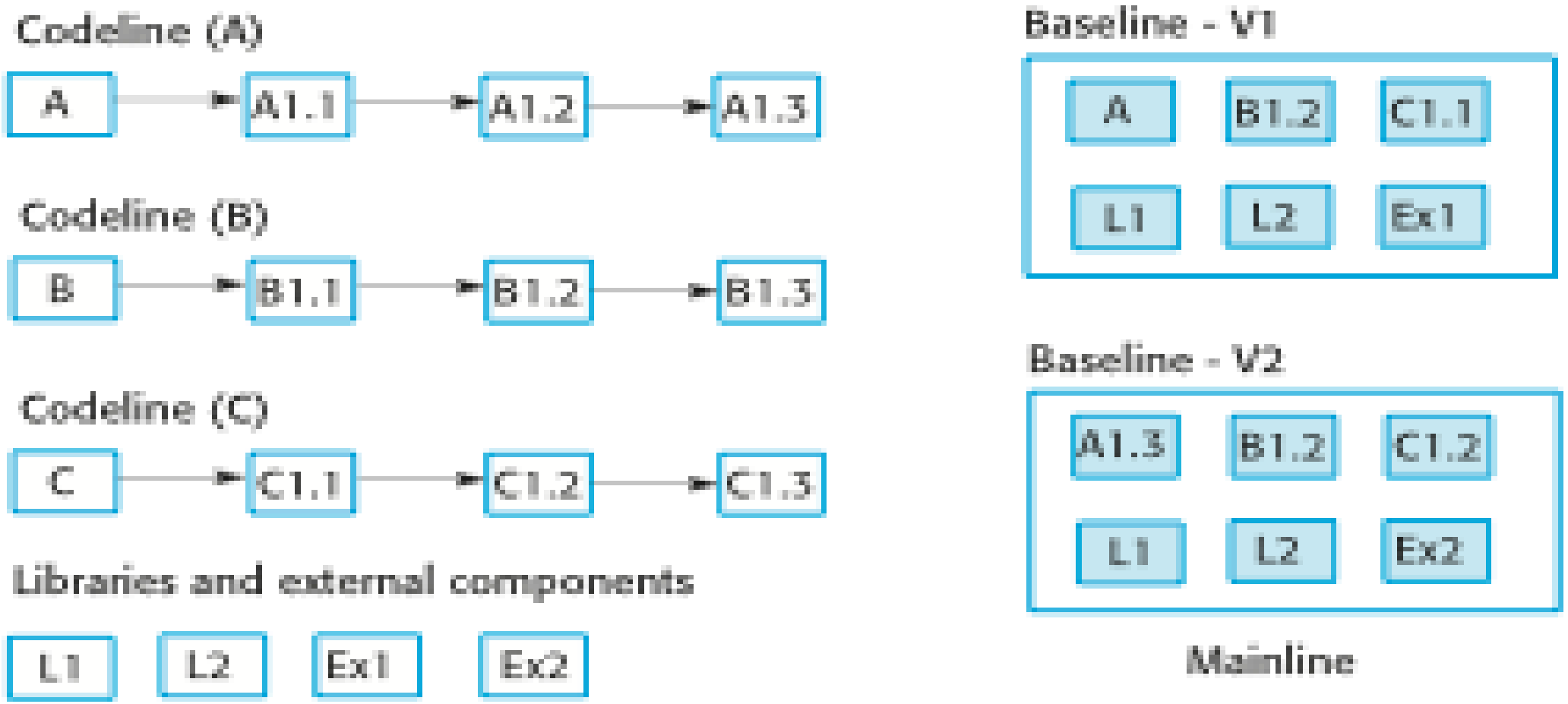
Configuration Management: Key Concepts

- A **Baseline** is a snapshot of the state of a product and any component products, frozen at a point in time for a particular purpose. For example, when the Project Plan is agreed between Project Manager and Project Board, it is 'baselined'.

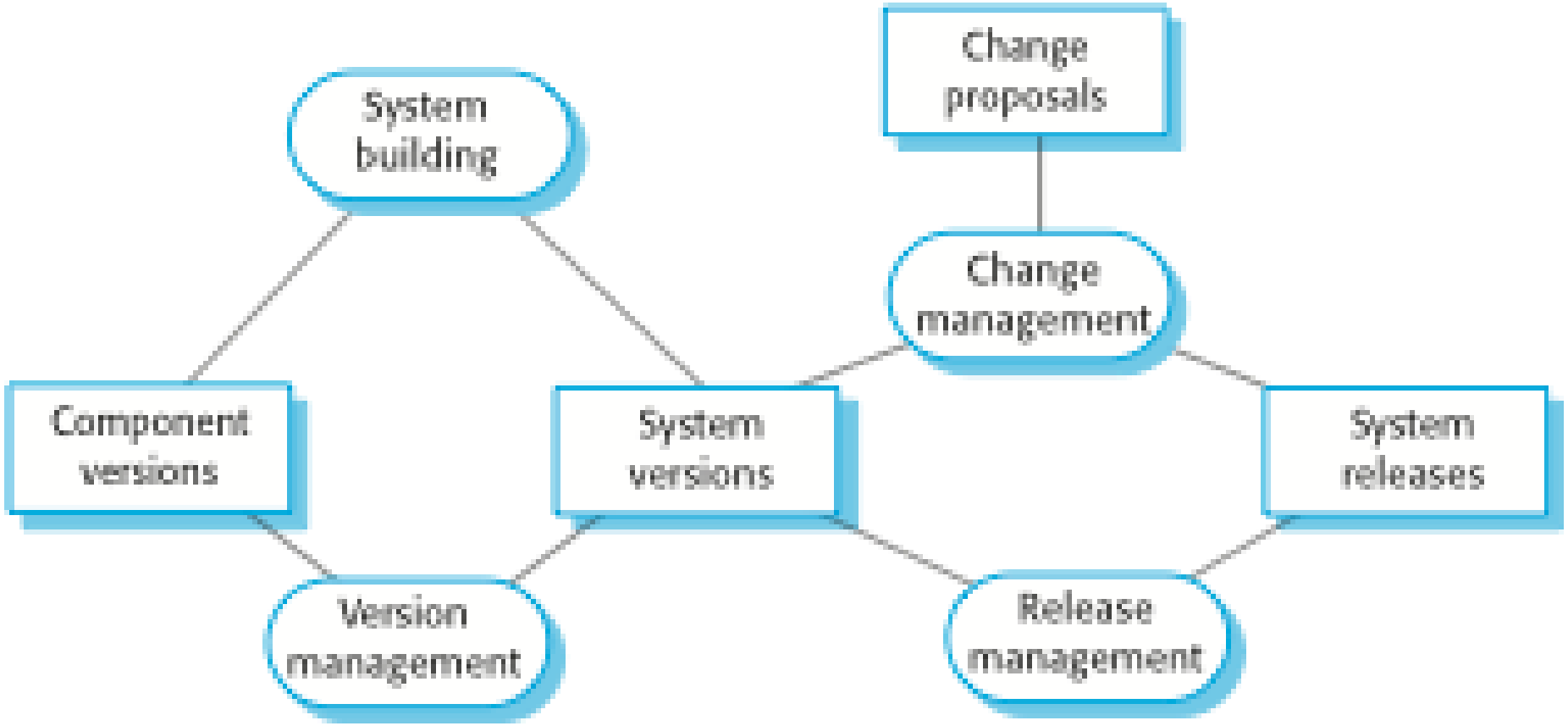
  In SW terms, a baseline is a frozen collection of component versions that make up a system. Baselines are controlled, which means that the versions of the components making up the system cannot be changed. This means that it should always be possible to recreate a baseline from its constituent components.
- **Change management:** Keeping track of requests for changes to the assets/products (e.g. software from customers and developers), working out the costs and impact of changes, deciding the changes should be implemented and track changes – (see next section)
- **System building:** The process of assembling program components, data and libraries, to create an executable system.
- **Release management:** A **release** is a complete and consistent set of products, which forms a fixed reference point in the development of the end outcome, e.g. a system version ready to become available to end-users. **Release Management** mainly concerns *planning*, *preparing* and *tracking* product versions for release.

Configuration Management: Key Concepts

Example of SW Codelines and Baselines



Configuration Management: Key Concepts



Change Management

Change Management: A process that manages change, i.e. "*the assessment of the impact of potential changes, their importance, their cost and a judgmental decision by management on whether to include them or not. Any approved changes must be reflected in any necessary corresponding change to schedule and budget.*" (Prince2)

Changes are associated to **Project Issues**, i.e. anything that could have an effect on the project. For example:

- Change in *Requirements*
- A change in the *Project Environment* such as: a legislative change, a new stakeholder, an unexpected change to a member of the project team, a programme management directive, a corporate reorganisation
- A *problem* occurring or being identified that was not anticipated during risk analysis
- *Problem* or *error* occurring in work completed or currently under way

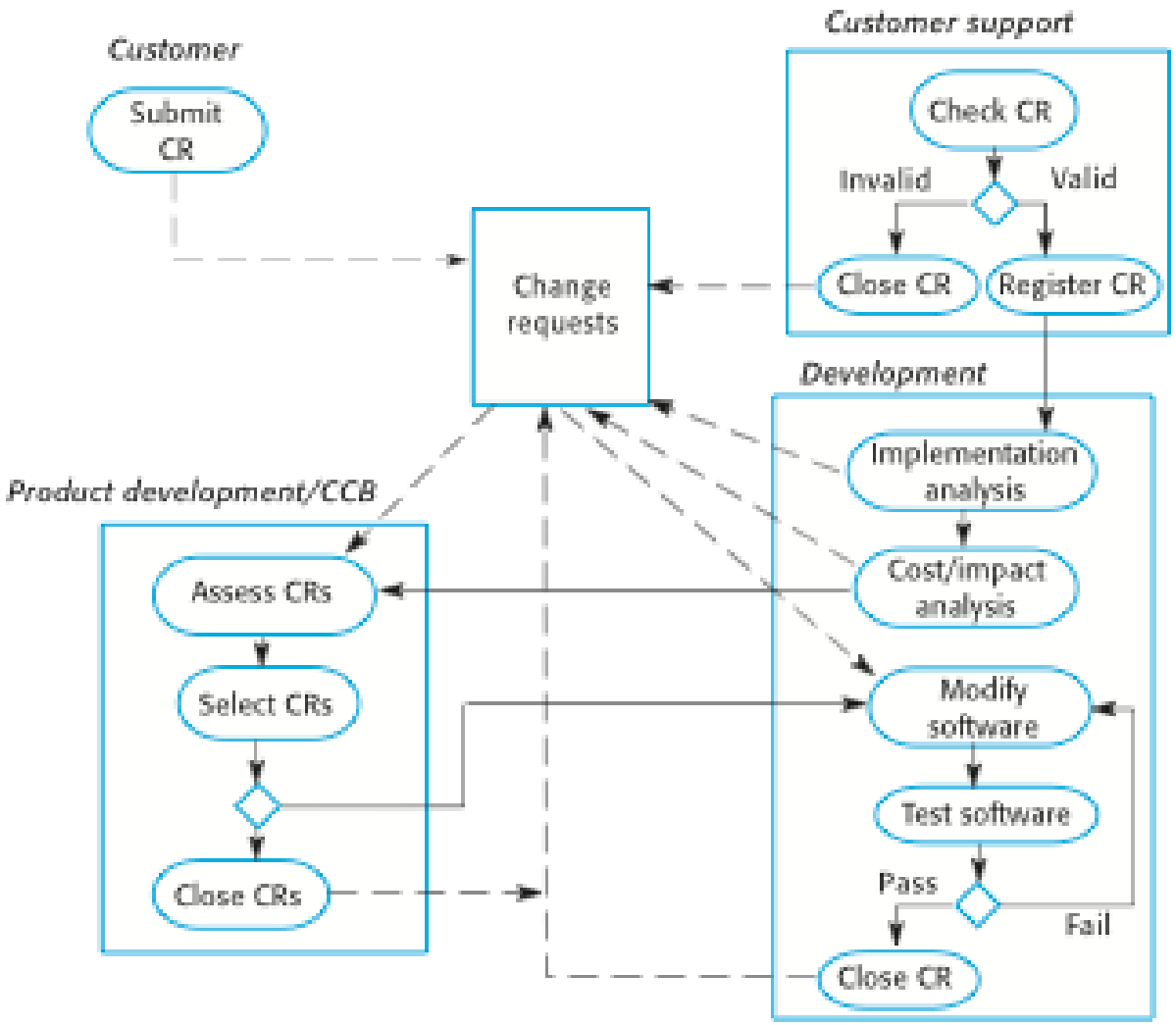
Project Issues are typically captured in an **Issue Log**, are assessed to determine/plan actions and reviewed to monitor progress. Issues are also related to Risks (How?)

Issues/Changes Assessment, Actions and Review are often performed by a **Change Control Board (CCB)** (e.g. the Project SC (Project Board) or to a delegate body). In Prince2 this is known as **"Change Authority".**

Changes formally originate via formal **Change Requests (CR).** The set of CRs are maintained in a relevant Log.

Change Management

Example Change Management Process (in SW product development)



Change Management

Example **Change Request Form**

Change Request Form		
Project Name:	**Submission Date:**	
Requested By:		
Description of Change		
Change Justification/Benefits		
Priority (H/M/L/Obligatory)**:**		
Impact:		
Impact on Cost, Schedule, Risk, Scope, Other		
Initial Evaluation	Name:	Evaluation Date:
Comments and Decision, Approvals Sign-Off		

Change Management

Example **Change Requests Log** (simplified)

Project Name					
Change Request Number	Change Request Description	Impact	Decision	Decision Body and Date	Comments

Change Management

Example **Project Issue Log** (simplified)

No	Issue	Date	Assigned To	Expected Resolution Date	Status (Open, In Progress, Closed)	Resolution Actions	Date Resolved	Comments

Note: Each organization/project typically has its own concrete change management process, with associated roles, responsibilities, tasks and document templates